Topic 1: Sets, Mappings, and the Language of Economic Models

(Why economists spend so much time talking about functions)

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1. Why this topic exists at all

When economists write down a model, they are almost never trying to compute a number.
They are trying to describe a **relationship**.

For example:

- how prices relate to demand,
- how preferences relate to choices,
- how information relates to outcomes,
- how today's actions relate to tomorrow's incentives.

Mathematically, a relationship is expressed by a **mapping**: something that takes inputs and assigns outputs.

Much of the difficulty students face with formal economic theory comes from not being fully comfortable with this basic ideas like:

- what are the objects?
- what is being mapped to what?
- what does it mean to "solve" the model?

This first topic is therefore about learning to **read and write the language in which economic relationships are expressed**.

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2. Sets: describing what can possibly happen

We begin with the simplest idea: a **set**.

A set is simply a collection of objects. These could be **things you own**, **options you have**, **things that can be produced**, etc.

If, for example, a consumer chooses quantities of two goods, she then has a set of goods. There are many ways to write them down, one we often use is a pair of numbers, say (x_1, x_2).
The set of all possible bundles is written as a subset of the two-dimensional plane $\mathbb{R}^2$.

Already here we see an important idea: **economic objects are often vectors, not scalars**.

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3. Cartesian products: building multidimensional spaces

Suppose we have two sets:

- X_1: possible choices of good 1,
- X_2: possible choices of good 2.

The **Cartesian product**

$$X_1 \times X_2$$

is the set of all ordered pairs (x_1, x_2) with $x_1 \in X_1$ and $x_2 \in X_2$.

Geometrically, you should think of a Cartesian product as a **multidimensional space**:

- one dimension per component,
- each point representing a complete specification of choices.

This construction appears everywhere in economics:

- a consumption bundle is a point in a product space,
- a strategy profile is a point in a product of players' strategy sets,
- a type profile is a point in a product of agents' type spaces.

Whenever you see notation like

$$x = (x_1, \ldots, x_n),$$

you are in a Cartesian product. Understanding this is essential for following later arguments about equilibrium and interaction.

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4. Functions: when behavior is uniquely determined

A **function** is a rule that assigns **exactly one output** to each input.[1]

Formally, a function f from a set X to a set Y is written as

$$f : X \to Y.$$

This formulation means **every** element in set X is mapped to **some** element in set Y, or formally $\forall x \in X, \exists y \in Y$ such that $f(x) = y$.[2] It does *not* mean: For every element y in Y there is (at least one) x such that $f(x) = y$.

In economics, functions are used when the model predicts a **unique response**.

Examples:

[1] Note that the opposite need not be true. So each output may have more than one input assigned to it. For more on this, see below when we talk about bijectivity and related concepts.

[2] You should read this subclause as: "... or formally *for each* ($\forall$) element x in set X *there exists* an element y in set Y such that plugging x into function f gives the result y. In terms of notation you may also sometimes see $x \mapsto f(x)$ when talking about specific elements rather than domain (X) and codomain (Y).

- a utility function assigns a number to each consumption bundle,
- a production function assigns output to inputs,
- an allocation rule in mechanism design assigns an outcome to reported types.

What matters is not just the formula, but the **direction of the mapping**:

- What is taken as given?
- What is determined as a consequence?

Economic reasoning is often about fixing the domain and studying how the output changes when the input changes.

5 Injective, surjective, and bijective mappings

So far, we have said what a function is, but not all functions behave in the same way.
In economics, it is often important to understand *how much information is lost or preserved* by a mapping.

Let $f : X \to Y$ be a function.

We say that f is **injective** (or one-to-one) if different inputs always lead to different outputs. Formally, this means:[3]

$$f(x) = f(x') \Rightarrow x = x'.$$

Intuitively, injectivity means that observing the output uniquely pins down the input. No two distinct elements of X are "collapsed" into the same element of Y.

In economic terms, injectivity is related to **identification**:

- if a mapping from types to outcomes is injective, observing the outcome reveals the type,
- if it is not injective, some information is necessarily lost (at least in some instances).

We say that f is **surjective** (or onto) if every element of Y is reached by *some* element of X. Formally:

$$\forall y \in Y, \exists x \in X \text{ such that } f(x) = y.$$

Surjectivity is about **coverage**:

[3] Read the following sentence like this: the value of function f with input x is equal to the value of function f with input x' **only if** x is the same as x'. This sounds overly complicated, but it also teaches us something about negations. Here, we would say "if x and x' are not exactly the same, then $f(x)$ and $f(x')$ *cannot* be the same when f is injective."

- the codomain Y does not contain "irrelevant" elements that $f(x)$ never reaches, because
- everything in Y can actually arise as an outcome if we pick the right x.

In economics, surjectivity often fails by design. For example, not every allocation may be feasible given technological or incentive constraints.

Finally, a function is **bijective** if it is both injective and surjective.

A bijection establishes a perfect one-to-one correspondence between X and Y:

- no information is lost,
- no outcomes are redundant.

Bijective mappings allow us to "relabel" objects without changing structure. They play an important role when we talk about equivalent representations, such as different utility functions representing the same preferences.

I should emphasize that **most economically interesting mappings are not bijective**. Demand is typically not injective, allocation rules are often not surjective, and equilibrium correspondences (see below) are almost never single-valued. Understanding this prevents a lot of confusion later on.

Counting with mappings: countable and uncountable sets

Injective and bijective functions (injections and bijections for short) also answer a question that sounds too simple to ask: how many elements does a set have?

For a small set we simply count. But look at what counting is. Saying that a set X has three elements means that we can label its elements "first, second, third" without skipping an element and without using a label twice. That is a bijection between X and $\{1, 2, 3\}$. Mappings let us carry the same idea over to sets that nobody could count to the end:

> Two sets have the same number of elements if there is a bijection between them. A set X has at most as many elements as a set Y if there is an injection from X into Y.

We will often care about the set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$. A set X is **countable** if there is an injective function $f : X \to \mathbb{N}$. It is **uncountable** if no such function exists.[4]

[4] Some books reserve "countable" for infinite sets of this kind and say "at most countable" for what we call countable. Nothing we do depends on the difference.

> "A set is countable if every element can be given its own natural number as a label. We can then write the elements down as a list, one after another, and every element appears at some finite position."

Which sets are countable?

- Every finite set, and $\mathbb{N}$ itself.
- The integers $\mathbb{Z}$: list them as $0, 1, -1, 2, -2, \ldots$
- The rational numbers $\mathbb{Q}$, that is, all fractions a/b with a an integer and b a natural number. This one is less obvious. Between any two fractions there is another fraction, so listing them in increasing order cannot work. Instead, sort them by the size of $|a|+b$: for each $k = 1, 2, 3, \ldots$ only finitely many fractions have $|a| + b = k$. Write down those with $k = 1$, then those with $k = 2$, and so on, skipping any number that is already on the list. Every rational number is reached after finitely many steps.

The real numbers $\mathbb{R}$ are **not** countable, and neither is any interval $[a, b]$ with $a < b$. No list, however cleverly arranged, contains every real number.[5]

Two facts are worth carrying along.

First, countability is inherited downwards. If Y is countable and there is an injection from X into Y, then X is countable as well: give each x the label of the element it is sent to. Different x are sent to different elements, and those carry different labels. In particular, every subset of a countable set is countable. Read the other way round, this is the form in which the concept usually does its work: **there is no injection from an uncountable set into a countable one.**

Second, $\mathbb{Q}$ is small when we count, but it is everywhere when we look at the real line. Between any two real numbers $a < b$ there is a rational number: pick a natural number n with $1/n < b - a$; the fractions with denominator n are spaced $1/n$ apart, so one of them has to fall strictly between a and b. We say that $\mathbb{Q}$ is **dense** in $\mathbb{R}$.

Why should an economist care? Recall that injectivity is about identification. Suppose an agent's type is a number in $[0, 1]$, say her willingness to pay, and she communicates by choosing a message. If messages are prices quoted in cents, or texts of finite length written in a finite alphabet, the set of messages is countable.[6] The set of types is not. So no rule that maps types to messages

[5] The argument is Cantor's. Suppose there were a list $x_1, x_2, x_3, \ldots$ containing every number in $[0, 1]$, and write each x_n as a decimal. Now build a number y digit by digit: its n-th digit after the decimal point is 5 if the n-th digit of x_n is not 5, and 6 otherwise. Then y lies in $[0, 1]$ but differs from every x_n in the n-th digit, so it is not on the list. (Using only the digits 5 and 6 takes care of the one subtlety, namely that $0.4999\ldots$ and $0.5000\ldots$ are the same number.) Since $t \mapsto a + t(b - a)$ is a bijection from $[0, 1]$ onto $[a, b]$, no interval $[a, b]$ can be listed either, and a list of $\mathbb{R}$ would contain a list of $[0, 1]$. Countability is not covered in MWG's Mathematical Appendix; see Ok, Chapter B.

[6] There are only finitely many texts of each length, so we can list all texts of length one, then all of length two, and so on — the same trick as for $\mathbb{Q}$.

can be injective: some distinct types necessarily send the same message, and whoever observes the message cannot tell them apart. No incentive problem is involved here. It is pure counting.

The same counting argument is behind a classical result in consumer theory. Topic 2, Section 7 notes that not every way of ranking bundles can be summarized by attaching a number to each bundle. In the standard example, the lexicographic ranking, such numbers would carve out uncountably many intervals of real numbers, no two of them overlapping. Since $\mathbb{Q}$ is dense, each of these intervals contains a rational number, and picking one from each interval would be an injection from an uncountable set into $\mathbb{Q}$.[7] Countable sets also return in Topic 7, where a function may have countably many kinks without harm, and in Topic 15, where probabilities add up over countably many events.

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6. Composing and undoing mappings

A mapping takes elements from one pool and maps on to elements in another pool. A composition of mappings does this in several steps. Such chains of mappings are relevant because sometimes we may labor into income, and income into bundles we buy, but the income in the middle is only relevant *because* it helps us buy bundles. The actual mapping we are interested in is labor into stuff we buy. And for that we need compositions.

Composing: doing one thing after another

If $f : X \to Y$ and $g : Y \to Z$, we can feed the output of f straight into g. The result is a new mapping from X to Z, written

$$g \circ f : X \to Z, \qquad (g \circ f)(x) := g\big(f(x)\big).$$

Read $g \circ f$ as g *after* f: the one written on the right happens first.[8]

Notice what the definition quietly demands: the codomain of f has to be a set that g accepts as an input. If f returns euros and g expects bundles, there is nothing to compose. This is the first place where the habit of asking *what lives where* pays for itself.

Economics composes mappings constantly, usually without saying so:

- **Indirect utility.** Demand $x(p, w)$ maps prices and wealth to a bundle, and utility u maps bundles to numbers. Composed, $v(p, w) := u\big(x(p, w)\big)$ is the best utility attainable at those prices and that wealth.
- **Cost.** A required output level maps to the cheapest input combination, and that combination maps to a bill.

[7] This is MWG, Example 3.C.1.

[8] This ordering confuses me evertime. It is this way round, however, because $(g \circ f)(x)$ ought to look like $g(f(x))$, and in $g(f(x))$ it is the f that sits next to the x.

In both cases the composed object is the one we actually care about, and the two pieces are how we compute it.

Undoing: the inverse of a mapping

Section 5 said that a bijection lets us "relabel" objects without changing structure. That relabelling is itself a mapping, and it deserves a name.

Let $f : X \to Y$ be a bijection. Its **inverse** $f^{-1} : Y \to X$ sends each y back to the unique x that produced it:

$$f^{-1}(y) = x \quad \text{exactly when} \quad f(x) = y.$$

Equivalently, in the notation we have just introduced,

$$f^{-1} \circ f = \mathrm{id}_X, \qquad f \circ f^{-1} = \mathrm{id}_Y,$$

where id is the identity mapping, which returns its input untouched. Do, then undo: nothing happened.

Now look back at Section 5 and notice why we need bijectivity:

- **Injectivity** is what makes $f^{-1}(y)$ *unambiguous*. If two different inputs had produced the same y, there would be no way to decide which one to send y back to, and f^{-1} would not be a function at all — it would assign two outputs to one input, which is precisely what Section 4 forbids.
- **Surjectivity** is what makes $f^{-1}(y)$ *available for every* $y \in Y$. If some y were never produced by anything, f^{-1} would have nothing to return there, and it would fail to be defined on all of its stated domain.

So "bijective" is not two unrelated requirements that happen to be bundled together. It is exactly the pair of conditions that turns "undoing f" into a function.

A warning about the symbol, because sometimes it is used in different contexts. We also write f^{-1} for the **preimage** of a set,

$$f^{-1}(B) := \{x \in X : f(x) \in B\},$$

and *that* makes sense for any f whatsoever, bijective or not.[9] Both will appear later—the preimage version is how continuity gets defined in Topic 14 and how measurability gets defined in Topic 15.

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[9] If f^{-1} is a function that's fine, if not we could still *define* something like an inverse—the preimage.

An economic case: inverse demand

Take a single good with demand $x(p)$, and suppose demand is strictly decreasing. Then $x(\cdot)$ is injective: different prices give different quantities. So it can be undone, and its inverse is the familiar **inverse demand** $p(x)$, the price at which the market would just absorb the quantity x.

Often, we write $x(p)$ and $p(x)$ interchangeably. It is worth thinking about it for a bit. They are the same information read in two directions, and the second direction exists only because for each price, we buy a different bundle.

Recall, that above, I claimed that most economically interesting mappings are not bijective, and gave demand as its example. Suppose demand is flat over a range of prices—the consumer buys the same quantity for a set of prices.[10] Then many prices produce one quantity, injectivity fails at that quantity, and there is no inverse demand *function* there. What survives is a *set* of prices consistent with that quantity, which is a correspondence (Section 7) rather than a function. It is still not hard to compute the inverse, but the inverse is not a function anymore.

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7. Correspondences: when economics refuses to choose uniquely

Very often, however, economic models do *not* deliver unique outcomes.

Consider a consumer who is indifferent between several bundles. Or a firm that has multiple profit-maximizing input choices.
Or a player who has several best responses.

In such cases, behavior cannot be described by a function. Instead, we use a **correspondence**.

A correspondence assigns **a set of outputs** to each input:

$$F : X \rightrightarrows Y.$$

This notation means:

> for each $x \in X$, $F(x)$ is a subset (not necessarily a single element) of Y. Other than that, correspondences behave *conceptually* very much like functions.[11]

Economically, looking at correspondences reflects an important modeling choice:

- non-uniqueness is not a bug,

[10] Think of, say, basic food like milk or so. If milk is a little more or less expensive, you won't adapt demand of it.

[11] Obviously, if f is a function it is also a correspondence. However if g is a correspondence, it may or may not be a function.

- it is a realistic feature of optimization and strategic interaction.

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8. Graphs: putting inputs and outputs together

To study functions and correspondences, it is often useful to look at their **graphs**.

Formally, the graph of a correspondence F is the set

$$\mathrm{Gr}(F) = \{(x, y) \in X \times Y \mid y \in F(x)\}.$$

Intuitively, the graph is the picture we would draw to represent the function. Geometrically, the graph lives in a higher-dimensional space that records both:

- the input,
- and all admissible outputs.

Later in the course, properties of graphs (such as being closed) will be crucial for proving:

- existence of equilibrium,
- stability of solutions,
- continuity of economic behavior.

For now, the key idea is simple with unidimensional functions:[12] **a graph makes the mapping visible**.

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9. Where this appears in Mas-Collel, Whinston, and Green (MWG henceforth).

The basic language of sets, functions, relations, and correspondences is developed in:

- **MWG, Mathematical Appendix A**

The appendix is concise and formal. These notes are meant to provide the *verbal intuition* that helps you read it fluently.

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10. An economic example: demand.

Consider a standard consumer problem:

$$\max_{x \in X} u(x) \quad \text{s.t. } p \cdot x \leq w.$$

[12] That is functions that map, e.g., from one number into another.

What is the solution?

The correct answer is not necessarily "a bundle that solves the problem." The correct answer is: "**the set of bundles* that solve the problem."

This set is called the **demand correspondence**:

$$x(p, w) \subseteq X.$$

Why is that the case? Because if p is not the price for a single good but a **price vector** capturing the prices of several goods, the consumer may be indifferent between very different bundles for some p and w, but as prices change slightly, she may not anymore. Thus, keeping track of those indifferences (and changes therein) is going to drive some results.

If you mistakenly think demand must be a function, many later results will seem mysterious or false.

What scaling does: homogeneity

The same example delivers a property of demand that we can read off without solving anything.

Multiply all prices and wealth by the same number, say 2. The constraint $2p \cdot x \leq 2w$ describes exactly the same bundles as $p \cdot x \leq w$: divide both sides by 2. The consumer chooses from the same set with the same u, so the set of solutions is the same. Nothing is special about the number 2. For every $\lambda > 0$,

$$x(\lambda p, \lambda w) = x(p, w).$$

> "Only ratios matter: the price of one good in terms of another, and wealth in terms of goods. Switching the currency unit from euros to cents multiplies every price and wealth by 100, and changes no choice."

This property has a name, and the name is used for mappings in general. A function f is **homogeneous of degree** r if, for every z in its domain and every $\lambda > 0$,[13]

$$f(\lambda z) = \lambda^r f(z).$$

In English:

> "Scaling the input by λ scales the output by λ^r."

Two degrees carry most of the weight in economics:

[13] The domain has to allow the question: together with every z it must contain λz for all $\lambda > 0$. The space $\mathbb{R}^n$ and the set of strictly positive price–wealth pairs are examples. For a correspondence of degree zero, the only case we need, the definition reads the same, with equality of sets.

- **Degree zero.** Since $\lambda^0 = 1$, the output does not react to scaling at all. Demand is homogeneous of degree zero in (p, w).
- **Degree one.** The output scales one-for-one. The cost $p \cdot x$ of a given bundle is homogeneous of degree one in prices: twice the prices, twice the bill. A production function with this property has *constant returns to scale*: doubling all inputs doubles output.

Once derivatives are available, homogeneity turns into an equation that ties all marginal effects of f to its level. That is Euler's formula in Topic 6.[14]

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11. Why this matters for microeconomics

Once you see economic models as **mappings between sets**, many things fall into place:

- equilibrium is a fixed point of a correspondence,
- incentive compatibility is a constraint on a mapping,
- equilibrium payoffs in repeated games form a set.

Microeconomic theory is, in many ways, the study of **structured relationships**. Learning to read and write these relationships carefully is the first step toward understanding everything that follows.

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12. What you should take away

After this topic, you should be comfortable with the following mental moves:

- asking what the domain and codomain of an object are,
- recognizing when outcomes are sets rather than points,
- visualizing multidimensional economic objects as product spaces.

You do **not** need to memorize formal definitions. You do need to develop the habit of *asking what lives where*.

[14] MWG covers homogeneous functions in Mathematical Appendix M.B and the homogeneity of demand in Section 2.E.